

ESPresSo++: A Fast and Extensible Molecular Simulation Package for Coarse-Grained Models

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Summary

ESPresSo++ is an open-source software package for molecular dynamics (MD) simulations with a particular emphasis on coarse-grained (CG) models of soft matter systems (Praprotnik et al., 2008). Written in C++ with a flexible Python interface, it is designed for high-performance computing (HPC) environments and supports massively parallel simulations through MPI. The package enables simulations of polymers, membranes, colloids and complex fluids with a wide range of interaction models and advanced algorithms.

ESPresSo++ builds upon the experience of its predecessor ESPResSo, but provides a cleaner, more modular codebase and enhanced extensibility. It is actively developed by an international community of researchers across multiple institutions and disciplines.

Statement of need

Molecular dynamics simulations are essential tools for exploring the behavior of soft matter systems at mesoscopic scales. General-purpose MD codes such as GROMACS (Abraham et al., 2015) and LAMMPS (Thompson et al., 2022) are primarily optimized for all-atom simulations and may require significant customization for coarse-grained models that need non-standard interactions or specialized multiscale algorithms. ESPResSo++ addresses this gap by providing:

- A modular and extensible design, enabling researchers to easily implement new interaction potentials and integrators.
- Efficient parallelization for large-scale simulations of complex systems.
- A rich library of coarse-grained interaction models and algorithms tailored to soft matter.
- A Python-based scripting interface for ease of use, reproducibility, and coupling with external analysis tools.
- Multi-scale simulation techniques such as AdResS and Lees-Edwards.

State of the field

Several established MD packages serve the soft matter and coarse-grained simulation communities. LAMMPS (Thompson et al., 2022) is a general-purpose, highly scalable code with broad force field support and a flexible plugin architecture; however, its input scripting language is less suited to complex CG workflows, and adaptive resolution is not a core feature. GROMACS (Abraham et al., 2015) provides exceptional performance for standard force

fields and supports the Martini CG model ecosystem, but adding truly novel CG interactions requires modifying the C++ source, and its file-based workflow is less interactive than a Python API. **HOOMD-blue** (Anderson et al., 2020) is the most directly comparable package, offering a Python-native API and GPU-accelerated CG simulations for soft matter; however, it lacks support for adaptive resolution methods. **OpenMM** (Eastman et al., 2017) excels at GPU-accelerated simulations and allows custom force definitions via algebraic expressions, but targets primarily biomolecular systems and lacks MPI-based multi-node parallelization. The sibling project **ESPResSo** (Weik et al., 2019) shares common roots but focuses on charged systems, hydrodynamic coupling (lattice Boltzmann), and electrokinetics rather than multiscale resolution bridging.

ESPResSo++ occupies a distinct niche as a package purpose-built for coarse-grained and multiscale soft matter simulations. Its key differentiator is first-class support for adaptive resolution simulation (AdResS), which allows seamless coupling of atomistic and coarse-grained representations within a single simulation. **ESPResSo++** is currently the only actively maintained MD package that provides AdResS as a core feature; none of the other packages listed above offer this capability. It is worth noting that these capabilities were also contributed to existing general-purpose codes (Fritsch et al., 2012; Junghans & Poblete, 2010; Nagarajan et al., 2013), but got removed again later, hence a dedicated package allows tighter integration of multiscale algorithms with the domain decomposition, neighbor list construction, and force computation pipeline — design choices that would be difficult to retrofit into architectures optimized for all-atom throughput.

Software design

ESPResSo++ uses a layered C++/Python architecture. The performance-critical computational kernel (force calculations, integration, domain decomposition, and communication) is implemented in C++17 and exposed to Python through Boost.Python bindings. Each Python-exposed C++ class provides a static `registerPython()` method, and a central registration dispatcher ensures that the full object hierarchy is available as the `espressopp` Python module. This design lets users assemble, configure, and control simulations entirely from Python while retaining the performance of compiled C++ for the inner loops.

Modularity through templates and extensions.

Interactions are implemented using C++ class templates parameterized by a potential type (e.g., `VerletListInteractionTemplate<Potential>`), so that adding a new pairwise potential requires only implementing the `computeEnergy()` and `computeForce()` methods of a `Potential` subclass. The integrator follows a similar pattern: an `MDIntegrator` base class exposes an `addExtension()` mechanism through which thermostats, barostats, constraints, and adaptive resolution layers are attached via Boost.Signals2 callbacks. This signal-based coupling keeps extensions decoupled from the integration loop and from each other.

Parallelization.

ESPResSo++ employs spatial domain decomposition with MPI. The simulation box is partitioned across MPI ranks using a `NodeGrid`, and each rank further subdivides its domain into linked cells via a `CellGrid`. Ghost particle exchange handles communication of boundary data. A non-blocking variant (`DomainDecompositionNonBlocking`) overlaps communication with computation, and the heterogeneous spatial domain decomposition algorithm (HeSpaDDA) (Guzman et al., 2017) provides load balancing for spatially inhomogeneous systems.

Performance optimizations.

Since version 2.0 (Guzman et al., 2019), **ESPResSo++** has been modernized with SIMD vectorization through a structure-of-arrays (SOA) particle data layout (`ParticleArray`) with 64-byte alignment, yielding an overall three-times speedup for short-range non-bonded force calculations (Vance et al., 2023). An improved cell decomposition scheme (Yao et al., 2004)

allows sub-decomposition into cells with a length of half or a third of the cutoff radius, reducing the number of unnecessary distance calculations.

Testing and documentation.

The code is tested through a combination of Boost.Test (C++) and Python unittest test suites, executed via CMake/CTest and continuous integration on GitHub Actions. User documentation is built with Sphinx; developer API documentation with Doxygen.

Research impact statement

ESPResSo++ has enabled research across a broad range of soft matter topics over more than a decade of active development. It has been used in numerous peer-reviewed studies, including investigations of:

- Polymer rheology and entanglement effects (Grommes et al., 2021, 2022, 2024, 2025; Grommes & Reith, 2020; Hsu & Kremer, 2020, 2023, 2024; Lee & Paul, 2020; Ohkuma et al., 2023; Singh et al., 2020; Tubiana et al., 2021; Zhao, Singh, et al., 2020)
- Polymer concepts in Polysomes organization (Kobayashi & Guzman, 2026)
- Lipid membranes, protein and vesicle dynamics (Bause & Bereau, 2021; Papež et al., 2023; Zhao, Cortes-Huerto, et al., 2020)
- Adaptive resolution simulations (Fiorentini et al., 2020; Thaler et al., 2020)
- Ionic liquids under shear flow (Gholami et al., 2025; Zhang et al., 2021)
- Coarse-grained conformational surface hopping (Rudzinski & Bereau, 2020)

The project is developed across multiple institutions — Max Planck Institute for Polymer Research, Max Planck Computing and Data Facility, Los Alamos National Laboratory, Johannes Gutenberg University of Mainz, and Heidelberg University — with contributions from both current and former developers documented in the AUTHORS file. The codebase has a public development history spanning more than ten years on GitHub, with continuous integration, code coverage tracking, and versioned releases.

Functionality

Key features of ESPResSo++ include:

- **Inter-particle interactions:** Lennard-Jones, Coulomb, soft repulsive, bonded interactions, tabulated potentials, and more.
- **Algorithms:** Molecular dynamics, Langevin dynamics, dissipative particle dynamics (DPD), Brownian dynamics, adaptive resolution simulations (AdResS), Monte Carlo sampling.
- **Electrostatics:** Particle–particle particle–mesh (P3M), Ewald summation, and other long-range methods.
- **Parallelization:** Domain decomposition using MPI, optimized for massively parallel architectures.
- **Python interface:** Full simulation control and analysis scripting in Python.
- **Extensibility:** Modular design allows easy addition of new force fields, integrators, or analysis tools.

New Features since ESPResSo++ v2.0

Since the last major release of ESPResSo++ v2.0 in 2018 (Guzman et al., 2019) a number of new functionalities and features have been added, including:

- **SIMD vectorization and related optimizations:** enhance compute performance on modern CPUs (Vance et al., 2023)

- 133 ▪ **Cell decomposition:** allow sub-decomposition into cells with a length of half or a third
- 134 of the cutoff for direct force calculations (Yao et al., 2004)
- 135 ▪ **HeSpaDDA:** heterogeneous spatial domain decomposition algorithm (HeSpaDDA) for
- 136 larger scale simulations (Guzman et al., 2017)
- 137 ▪ **new potentials and simulation methods:** AngularCosineSquared, TabulatedSubEnsAngular,
- 138 surface hopping MD, Lees-Edwards boundary conditions
- 139 ▪ **Checkpoint the state of the random number generator (RNG):** allow restarting from
- 140 checkpointed state of RNG
- 141 ▪ **I/O:** support for parallel writing and reading of H5MD checkpoints
- 142 ▪ **Python 3 compatibility**

143 Example usage

144 A minimal Python script to run a simple (repulsive only) Lennard-Jones type particle simulation
 145 in ESPResSo++ looks like:

```
import espressopp
from mpi4py import MPI

# simulation system parameters
num_particles = 10000      # total number of particles in the system
box           = (20,20,20) # size of the simulationbox (all length are in sigma)
rc            = 1.12246    # cut off for the short range non bonded potential
skin         = 0.3        # skin used for verlet neighbor list
dt           = 0.005      # time step for 1 md step
epsilon      = 1.0        # energy unit
sigma        = 1.0        # length unit
temperature  = 1.0        # temperature of the simulation
LJcaprad     = 0.8        # initial capping radius for LJ interaction
                        # for random configurations

# system setup
system       = espressopp.System( )
system.rng   = espressopp.esutil.RNG( )
system.bc    = espressopp.bc.OrthorhombicBC(system.rng, box)
system.skin  = skin

# define underlying storage system for parallelisation
nodeGrid     = espressopp.tools.decomp.nodeGrid(MPI.COMM_WORLD.size,box,rc,skin)
cellGrid     = espressopp.tools.decomp.cellGrid(box, nodeGrid, rc, skin)
system.storage = espressopp.storage.DomainDecomposition(system, nodeGrid, cellGrid)

# interaction setup, here short range non-bonded Lennard Jones potential
interaction   = espressopp.interaction.VerletListLennardJonesCapped(
                    espressopp.VerletList(system, cutoff=rc))
interaction.setPotential(type1=0, type2=0,
                        potential=espressopp.interaction.LennardJonesCapped(
                            epsilon, sigma, shift='auto', caprad=LJcaprad))
system.addInteraction(interaction)

# integrator setup
integrator    = espressopp.integrator.VelocityVerlet(system)

# thermostat setup
thermostat    = espressopp.integrator.LangevinThermostat(system)
```

```

thermostat.gamma          = 1.0
thermostat.temperature = temperature
integrator.addExtension(thermostat)

# create random particle setup in the simulation box
props = ['id', 'type', 'mass', 'pos', 'v']
new_particles = []
pid = 1
while pid <= num_particles:
    type = 0
    mass = 1.0
    pos = system.bc.getRandomPos()
    vel = espressopp.Real3D(0.0, 0.0, 0.0)
    part = [pid, type, mass, pos, vel]
    new_particles.append(part)
    if pid % 1000 == 0:
        system.storage.addParticles(new_particles, *props)
        system.storage.decompose()
        new_particles = []
    pid += 1
system.storage.addParticles(new_particles, *props)

integrator.dt = 0.0001 # very small timestep for initial warmup
for n in range(20):
    # warmup finished, switch to uncapped Lennard Jones potential and increase timestep dt
    if n == 10:
        interaction = espressopp.interaction.VerletListLennardJones(
            espressopp.VerletList(system, cutoff=rc))
        interaction.setPotential(type1=0, type2=0,
                                potential=espressopp.interaction.LennardJones(
                                    epsilon, sigma, rc, shift='auto'))
        system.removeInteraction(0)
        system.addInteraction(interaction)
        integrator.dt = dt
        integrator.run(10000)
        Etot = system.getInteraction(0).computeEnergy()
        print("md time = {:.41f}, total energy: {:.10.2f}".format(integrator.dt*n*10000, Etot))

# write PDB file of (quasi) equilibrated LJ system.
# At this temperature it is more or less crystallized.
espressopp.tools.pdbwrite("simplelj.pdb", system, molsize=num_particles)

```

146 AI usage disclosure

147 No generative AI tools were used in the creation of the ESPReso++ software or its
 148 documentation. During the preparation of this manuscript, AI-assisted tools were used
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